The Geography of the Attack

Where points are actually won in the 2025-2026 Serie A2 volleyball championship

Data Visualization Lab — Project Proposal (Deliverable 1)

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1. Project Idea Description

1.1 Topic

Every volleyball coach has an idea of how a team wins points. This idea is developed through experience, technical knowledge, training, and the interaction between players throughout the season. During competitions, teams collect large amounts of scouting data that describe their playing style and tactical choices. These data are mainly used by coaches to analyze opponents, while they are less often used to evaluate and improve their own team's attacking patterns. This project investigates whether specific characteristics of an attack, such as the attack zone, tempo, and number of blockers faced, can predict the probability of scoring a point.

The analysis is based on rally-level DataVolley scouting data collected during the entire 2025/26 Serie A2 season for two teams. The dataset includes 63 matches against around thirteen different opponents. Since each rally can contain multiple attacking actions, attacks are the most frequently recorded offensive skill in the dataset. This provides enough observations to analyze the combined effect of attack zone, tempo, and blocking situation.

The results may be useful for coaches and performance analysts by providing evidence about which attacking situations are more effective. They may also help players and volleyball fans better understand the tactical factors that contribute to winning rallies, beyond the final score.

The two teams considered in this study, Brescia and Talmassons, both achieved promotion from Serie A2 during the 2025/26 season. This makes the comparison particularly interesting. Instead of simply comparing which team was stronger, the project investigates whether they achieved the same result using similar attacking strategies or different tactical approaches. Talmassons earned direct promotion, while Brescia reached Serie A1 through the playoffs, playing six additional matches. Despite these different paths, the two teams finished the regular season separated by only one point, making them a suitable case study for comparing successful attacking strategies.

1.2 Circumstances

1.2.1 Target Audience

The main audience for this project is volleyball coaches and performance analysts, who are already familiar with concepts such as attack zones (4, 3, 2, and back row), attack tempo, and the number of blockers. A second audience includes players and volleyball fans who follow Serie A2 and understand the basic dynamics of attack and block, even if they are not familiar with DataVolley terminology. For this reason, the visualization uses a simple volleyball court and net diagram instead of technical scouting codes, making the results easy to understand for both groups.

1.2.2 Format and Medium

The final output will be an interactive web application developed with Bokeh. The main visualization is based on a volleyball court and net diagram, where users can move the cursor over each attack zone to see statistics such as kill rate, error rate, and block rate. Interactive filters allow users to select the team, attack tempo, and number of blockers, updating all the visualizations at the same time.

Bokeh was chosen because it is the interactive visualization library used during the course and is well suited to this type of project. During the proposal stage, the visualizations are presented as static figures created with Matplotlib and Seaborn. These will later be converted into interactive Bokeh visualizations that can be explored on a laptop or tablet.

1.3 Purpose

The main goal of the visualization is to explain how attacking effectiveness changes according to the attack zone, tempo, and blocking situation. It aims to show which combinations are more likely to produce a point and which are less effective than expected.

A second goal is to compare different attacking situations and the two teams included in the study. The visualization allows users to explore which attacking choices are more successful, how performance changes depending on the number of blockers, and whether Brescia and Talmassons achieved promotion using similar or different attacking strategies.

After exploring the visualization, users should understand that attacking effectiveness cannot be explained only by the attack zone alone. Instead, it is influenced by several factors, especially attack tempo and the number of blockers. They should also be able to identify specific situations in which the data supports or differs from common coaching expectations.

2. Project Data

The dataset consists of DataVolley scouting files from 63 Serie A2 matches played during the 2025/26 season. It includes complete scouting data for two teams: 29 matches for Talmassons and 36 for Brescia, including the two matches played against each other. Each file contains rally-by-rally information, such as the skill performed (serve, reception, set, attack, block, dig, and freeball), attack zone, attack tempo, attack evaluation, hitter, setter, number of blockers, team rotation, and the current score.

These data are suitable for the project because attack is the most frequent offensive action recorded during a match. While each rally contains only one serve, it may include several attacking actions, such as the first attack, transition attacks, back-row attacks, or attacks after a defensive dig. This provides a large number of observations, making it possible to analyze attacking performance according to attack zone, tempo, and number of blockers. In addition, the two teams faced around fifteen to eighteen different opponents during the season, making the analysis more representative than a comparison based on only a few matches.

The data are not publicly available, as DataVolley scouting files are produced and owned by each team's coaching staff. Before the analysis, the files must be exported from the DataVolley format and cleaned by standardizing variables such as attack zones, tempos, and evaluation codes.

In practice, the raw scouting files are not immediately usable for analysis: each action is stored as a compact code (team, player, skill, evaluation, and a set of trailing detail fields) rather than as a set of labeled columns. Understanding how to interpret this format benefited from guidance provided by the team's scout analyst, who clarified the scouting conventions used during data collection. Turning this raw format into a clean, analysis-ready dataset is done using the pydatavolley library (openvolley/pydatavolley), which decodes the raw action codes into structured fields for all 63 files.

One field needed particular care before it could be used: pydatavolley decodes the number of opposing blockers from a fixed character position in the action code, but a fourth, physically impossible value also appears in the data (no real attack can face 4 front-row blockers). Establishing whether this was scouting noise or a real, systematic pattern — rather than assuming either — was necessary before treating the field as usable; the Technical Report (Section 4) documents that investigation and confirms hypotheses 2 and 3 below, both of which depend on a reliable blocker count, are indeed testable with the available data. The same check is included as a runnable code cell in the parsing step of the project's notebook, so the result can be reproduced directly rather than taken on faith.

Some limitations should also be considered. Since the scouting is performed by human operators, small differences in coding may exist, especially when classifying attack tempo. Moreover, attack effectiveness depends on several factors, including the quality of the set and the opponent's block, not only on the attacker. Finally, when the data are divided by attack zone, tempo, and number of blockers at the same time, some combinations may contain relatively few observations and may need to be grouped together during the analysis.

3. Project Definition

3.1 Potential Views and Visual Encodings

- **Court/net heatmap (Seaborn):** a volleyball court diagram where each attack zone is colored according to its kill rate, error rate, or block rate. This will be the main visualization of the project, as it allows users to relate performance to the position on the court immediately.
- **Cleveland dot plot or diverging bar chart:** used to rank zone-tempo combinations according to their net efficiency (kill rate − error rate). These visualizations make it easier to compare many categories than a standard grouped bar chart and clearly highlight the most and least effective attacking situations.
- **Seaborn jointplot:** a scatter plot with marginal distributions to explore the relationship between variables such as set quality and attack outcome. This visualization also shows how many observations support the results, helping to avoid misleading conclusions based on very small samples.
- **FacetGrid small multiples:** the same heatmaps and dot plots are displayed separately for Brescia and Talmassons, allowing an easy comparison between the two teams. The same approach can also be used to compare the regular season with the playoffs.

- **Stretch goal – PCA on attacking profiles:** each match or player can be represented by variables such as attack zone distribution, attack tempo distribution, and efficiency. PCA would project these profiles into two dimensions to explore whether different attacking styles emerge.

The main hypotheses are:

1. Quick middle attacks remain the most effective attacking option across different blocking situations, while high-ball attacks are expected to be the least effective.
2. Back-row attacks perform well against a single blocker but lose effectiveness against a well-formed block.
3. The effectiveness of an attack zone depends not only on the zone itself, but also on the number of blockers. For this reason, the visualizations will allow users to filter the results by blocking situation instead of showing a single overall ranking.

3.2 Data Requirements

The visualizations require the following variables: skill type, attack zone, attack tempo, attack evaluation, number of opposing blockers, hitter, setter, rotation, and the current score. One additional variable will be created by grouping DataVolley's detailed tempo codes into four simpler categories: quick, medium, high-ball, and pipe. This simplifies the visualization while preserving the main tactical differences.

To avoid unreliable results, combinations of attack zone, tempo, and number of blockers with very few observations will either be grouped with similar categories or clearly identified. Opponent identity will also be available as a filter, since blocking quality depends on the opposing team. Keeping this information allows a more accurate interpretation of the results and avoids attributing attack effectiveness only to the attack zone or tempo.

3.3 Highlighting Key Features

The same volleyball court diagram will be used throughout the project to provide a consistent reference across all visualizations. A diverging color scale will distinguish more effective from less effective attack situations, allowing users to identify the main patterns quickly.

In the final Bokeh application, users will be able to move the cursor over each attack zone to view detailed statistics, including the exact values and the number of observations. The different visualizations will also be linked, so selecting a zone in one view will highlight the corresponding information in the others. This will make it easier to explore the data without displaying every detail at the same time.

Finally, attack zone and tempo combinations that differ from common coaching expectations will be highlighted to draw attention to the most interesting findings.

4. Expected Challenges

- **Data quality and consistency:** scouting is performed by different analysts, so attack tempo and zone may not always be coded in the same way. The data will therefore be checked and standardized before comparing the two teams.
- **Data parsing:** the DataVolley files must be processed to correctly reconstruct each rally and identify the phase of play (first-ball attack or transition attack) together with the blocking situation.
- **Confounding factors:** attack effectiveness depends not only on attack zone and tempo, but also on factors such as set quality and hitter ability. These variables should be considered during the analysis to avoid misleading conclusions.
- **Sample size:** some combinations of attack zone, tempo, and number of blockers may contain only a few observations. Similar categories may need to be grouped to ensure reliable results.
- **Learning curve:** Bokeh and PCA are new tools compared with those used in previous labs. For this reason, static visualizations created with Matplotlib and Seaborn will be developed first and kept as a backup if the interactive visualizations or PCA cannot be completed within the project timeline.